Biometrical Modelling of Nuclear Family Trio Data Using Linear Mixed Models (AE / ACE / ADE)

Contents

1 Introduction

We present a unified framework for estimating genetic and environmental variance components from **nuclear family trios** (motherfatherchild) using standard linear mixed models.

The approach follows the key idea of Rabe-Hesketh, Skrondal & Gjessing (2008): pedigree likelihoods can be written as mixed models by exploiting covariance equivalence.

This allows biometric models to be fitted using standard software (`nlme` in R) rather than specialised SEM packages.

2 Quantitative Genetic Models

2.1 ACE model

$$y_{ij} = X_{ij}\beta + a_{ij} + c_j + e_{ij}$$

$$a_{ij} \sim N(0, \sigma_A^2), \qquad c_j \sim N(0, \sigma_C^2), \qquad e_{ij} \sim N(0, \sigma_E^2)$$

2.2 ADE model

$$y_{ij} = X_{ij}\beta + a_{ij} + d_{ij} + e_{ij}$$

$$a_{ij} \sim N(0, \sigma_A^2), \qquad d_{ij} \sim N(0, \sigma_D^2), \qquad e_{ij} \sim N(0, \sigma_E^2)$$

Total phenotypic variance:

$$\sigma_P^2 = \sigma_A^2 + \sigma_C^2 + \sigma_D^2 + \sigma_E^2$$

Only a subset of these components can be identified from trio data.

3 Pedigree Covariance in Nuclear Families

The covariance between relatives i and k is

$$\mathrm{Cov}(y_i, y_k) = 2\phi_{ik}\sigma_A^2 + \sigma_C^2$$

3.1 Kinship Coefficients

Pair	2ϕ	Covariance
Mother–Father	0	σ_C^2
Mother–Child	$\frac{1}{2}$	$\frac{1}{2}\sigma_A^2 + \sigma_C^2$
Father–Child	$\frac{1}{2}$	$\frac{1}{2}\sigma_A^2 + \sigma_C^2$

3.2 Dominance covariance in trios

Dominance effects require two alleles identical by descent. Parent–child pairs do not share dominance deviations, therefore

$$\mathrm{Cov}_D(y_i, y_k) = 0 \quad \text{for all pairs in trios.}$$

Dominance contributes only to individual variance:

$$Var(y) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2$$

4 Mixed Model Representation

$$y = X\beta + Z_A u_A + Z_C u_C + Z_D u_D + e$$

$$u_A \sim N(0, \sigma_A^2 I), \quad u_C \sim N(0, \sigma_C^2 I), \quad u_D \sim N(0, \sigma_D^2 I)$$

5 Trio Design Matrix Construction

Dataset indicators:

$$\text{var1} = \text{mother}, \quad \text{var2} = \text{father}, \quad \text{var3} = \text{child}$$

Additive genetic contribution:

$$A_{ij} = 0.5\,\text{Mother} + 0.5\,\text{Father} + 1\,\text{Child}$$

Shared family environment:

$$C_{ij} = 1$$

Dominance deviation (individual-level):

$$D_{ij} = 1$$

6 Identifiability

With nuclear family trios:

Model	Identifiable?
AE	Yes
ACE	Yes
ADE	Yes
ACDE	No

Shared environment and dominance cannot be separated simultaneously, so ACE and ADE are mutually exclusive.

7 Variance Components

7.1 AE model

$$\Sigma = \sigma_A^2 K + \sigma_E^2 I \qquad h^2 = \frac{\sigma_A^2}{\sigma_A^2 + \sigma_E^2}$$

7.2 ACE model

$$\Sigma = \sigma_A^2 K + \sigma_C^2 J + \sigma_E^2 I$$

$$h^2 = \frac{\sigma_A^2}{\sigma_P^2}, \qquad c^2 = \frac{\sigma_C^2}{\sigma_P^2}$$

7.3 ADE model

$$\Sigma = \sigma_A^2 K + \sigma_D^2 I + \sigma_E^2 I$$

$$h^2 = \frac{\sigma_A^2}{\sigma_P^2}, \qquad d^2 = \frac{\sigma_D^2}{\sigma_P^2}$$

8 Likelihood Model Comparison

$$LR = 2(\ell_{\text{full}} - \ell_{\text{reduced}})$$

9 R Implementation

9.1 Unified Trio Model Function (AE / ACE / ADE)

```
trio_model <- function(model, data, type=c("AE","ACE","ADE"))
{
  require(nlme)
  type <- match.arg(type)

  data$Acoef <- 0.5*data$var1 + 0.5*data$var2 + 1*data$var3
  data$Ccoef <- 1
  data$Dcoef <- 1
  data$indid <- 1:nrow(data)

  if(type=="AE")
    rand <- list(familyid = pdDiag(~ Acoef - 1))

  if(type=="ACE")
    rand <- list(familyid = pdDiag(~ Acoef + Ccoef - 1))

  if(type=="ADE")
    rand <- list(
      familyid = pdDiag(~ Acoef - 1),
      indid    = pdDiag(~ Dcoef - 1)
    )

  fit <- lme(model, random=rand, data=data, method="ML",
             control=lmeControl(opt="optim"))

  pars <- attr(fit$apVar,"Pars")

  varE <- exp(2*pars[1])
  varA <- exp(2*pars[2])
  varC <- if(type=="ACE") exp(2*pars[3]) else 0
  varD <- if(type=="ADE") exp(2*pars[3]) else 0

  varP <- varA + varC + varD + varE
```

```
  list(
    fit = fit,
    var = c(A=varA,C=varC,D=varD,E=varE),
    h2  = varA/varP,
    c2  = varC/varP,
    d2  = varD/varP
  )
}
```

9.2 Model Comparison

```
compare_trio_models <- function(model,data)
{
  AE  <- trio_model(model,data,"AE")
  ACE <- trio_model(model,data,"ACE")
  ADE <- trio_model(model,data,"ADE")

  tab <- anova(AE$fit, ACE$fit, ADE$fit)

  list(AE=AE, ACE=ACE, ADE=ADE, anova=tab)
}
```

10 Example

```
library(gap.datasets)
data(mfblong)

model <- bwt ~ male + first + midage + highage + birthyr
res <- compare_trio_models(model, mfblong)

res$anova
res$ACE$h2
res$ADE$d2
```

11 Conclusion

Nuclear family biometric models can be estimated entirely within linear mixed models. Trio data allow estimation of AE, ACE or ADE models, but shared environment and dominance cannot be estimated simultaneously.